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# Title: Implement Stack using arrays.

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#include <stdio.h>
#define MAX 5

int stack[MAX];
int top = -1;

void push(int val){
   if(top == MAX - 1){
      printf("Stack Overflow!\n");
   }
   else{
      top++;
      stack[top] = val;
      printf("Pushed: %d\n", val);
   }
}

void pop(){
   if(top == -1){
      printf("Stack Underflow!\n");
   }
   else{
      printf("Popped element: %d\n", stack[top]);
      top--;
   }
}

void display(){
   int i;
   if(top == -1){
      printf("Stack is empty\n");
   }
   else{
      printf("Stack elements: ");
      for(i = top; i >= 0; i--){
         printf("%d ", stack[i]);
```

```
        }
        printf("\nTop position: %d\n", top);
    }
}

int main(){
    int choice, val;

    while(1){
        printf("\n--- STACK MENU ---\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        if(choice == 1){
            printf("Enter value: ");
            scanf("%d", &val);
            push(val);
        }
        else if(choice == 2){
            pop();
        }
        else if(choice == 3){
            display();
        }
        else if(choice == 4){
            break;
        }
        else{
            printf("Invalid choice!\n");
        }
    }

    return 0;
}
```